Even Hypercubes Decompose into Paths of Every Admissible Length

https://github.com/emzeay/erde-hypercube-paths

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Abstract

We prove that, for every even integer $n \geq 2$, the edges of the hypercube Q_n can be partitioned into simple paths of any prescribed length ℓ satisfying $1 \leq \ell < 2^n$ and $\ell \mid n2^{n-1}$, resolving the conjecture proposed by Joshua Erde. Subdivision reduces the problem to constructing a decomposition at the largest admissible length for each odd divisor of n. Building on Hamiltonian decompositions, representing sets, anchored products, and alternating anchored circuits, we construct these decompositions uniformly in the dimension and the odd divisor. Binary address arguments assign representatives and control the multiplicities of marked vertices. Noncrossing pairings coordinate the attachment of further coordinate blocks, producing pairs of simple cycles of the required length. An edge exchange opens these cycles into paths while preserving their edge partition. We also determine the minimum number of equal-length paths and transfer the decomposition theorem to Cartesian products of graphs admitting perfect-matching factorizations.

1 Introduction

In 2014, Joshua Erde [1, Conjecture 8] proposed the following conjecture about decomposing the edges of even-dimensional hypercubes into paths of equal length:

For every even integer $n \geq 2$ and every integer $\ell \geq 1$ satisfying

$$\ell < 2^n \qquad \text{and} \qquad \ell \mid n2^{n-1},$$

the edges of the hypercube Q_n can be partitioned into simple paths, each having exactly ℓ edges. Different paths may share vertices.

An even-dimensional hypercube has even dimension n, such as 2, 4, or 6. Erde settled the odd-dimensional case, where the necessary and sufficient conditions are $\ell \mid n2^{n-1}$ and $\ell \leq n$ [1, Theorem 2]. The conjecture above says that, for any even-dimensional cube, we can split all its edges into simple paths (routes along edges that never revisit a vertex) of any chosen admissible length.

The length of a path is its number of edges. An admissible length is a positive integer ℓ that is smaller than the number of vertices (2^n) and divides the total number of edges ($n2^{n-1}$). The first restriction bounds the number of vertices a simple path can visit. The second ensures that the total edge count divides evenly into pieces of the chosen size.

Every admissible length means that each allowed choice of ℓ has its own decomposition, with all paths in that decomposition having length ℓ. Every edge must be used exactly once. Different paths may meet at vertices but may not share edges.

For even dimensions, Axenovich, Offner, and Tompkins constructed decompositions into long cycles and proved that every admissible path length $\ell \leq 2^n/n$ is attainable [3, Theorems 1 and 2]. Their approach uses representing sets and anchored products to combine constructions on smaller coordinate blocks. Their cycle decompositions also yield path lengths beyond this general bound by subdivision.

The proof presented here builds on these representing sets and anchored products, the alternating anchored circuit approach described in the SUAMI research poster [6], and an established edge exchange that opens pairs of cycles into paths [5, Construction 1]. In the higher-weight cases, we coordinate shared endpoints and marked vertices across the factors. Binary address arguments control the number of marked appearances of each vertex for arbitrary odd parts. The resulting constructions yield paths of every admissible length. Section 5 derives consequences for optimal path counts, omitted vertex sets, and decompositions of other Cartesian products.

2 Proof overview

We first reduce the number of path lengths we need to construct. Write the requested length as $\ell = d\,2^a$, where d is odd. The divisibility condition $\ell \mid n2^{n-1}$ implies that $d \mid n2^{n-1}$. Since d is odd, it is coprime to 2^{n-1}, and so $d \mid n$. As n is even and d is odd, the quotient n/d is even, and we can write $n = 2md$ for some positive integer m. Choose r so that $2^r \leq d < 2^{r+1}$. It is enough to construct a decomposition into paths of length

$$M_d = d\,2^{n-r-1}$$

which is the largest admissible length with odd part d. Every shorter admissible length with the same odd part divides M_d, so we can obtain it by cutting each long path into equal consecutive pieces. Cutting a simple path preserves simplicity and does not lose or duplicate any edges.

It remains to construct the maximal-length paths for each positive odd divisor d of n. We divide these odd parts into the four families in the table below. The binary weight of a number is the number of ones in its binary expansion. A Mersenne number has a binary expansion consisting entirely of ones. Equivalently, it has the form $2^q - 1$ for a positive integer q.

Family of odd parts d	**Form**
One	$d = 1$
Two binary ones	$d = 2^r + 1$, with $r \geq 1$
Higher binary weight, non-Mersenne	At least three ones, with a zero somewhere between the highest and lowest bits
Higher binary weight, Mersenne	$d = 2^{r+1} - 1$, with $r \geq 2$

These families are disjoint and exhaustive. Every positive odd integer has a one in its lowest binary position. If that is its only one, then $d = 1$. If it has exactly two ones in its binary representation, then it is a power of two plus one. Otherwise, it has at least three ones. Its binary representation either consists entirely of ones from the highest bit down to the lowest bit, or has a

zero somewhere between the highest and lowest bits. Thus every positive odd divisor d belongs to exactly one of these four families.

We outline the construction for each family below. Section 3 gives the detailed proofs.

One. When $d = 1$, the required maximal length is $M_1 = 2^{n-1}$. An even-dimensional cube has a Hamiltonian decomposition [2], so its edges partition into cycles that each visit every vertex once before returning to the start. Each Hamiltonian cycle has length 2^n. Splitting its edge sequence into two equal halves gives two simple paths of length 2^{n-1}, which proves this case.

Two binary ones. For this family, we use the representing sets, anchored products, and wiggle construction of Axenovich, Offner, and Tompkins [3]. When $d = 2^r + 1$, write $P = 2^r$ and $n = 2m(P + 1)$. Split the cube into coordinate blocks of dimensions $2mP$ and $2m$. Repeated applications of the square construction to the smaller block's Hamiltonian decomposition give a Hamiltonian decomposition of the larger block. On each cycle, we select equally spaced vertices forming a representing set. We call these vertices representatives. The square construction lets us choose these sets so that, within each group of descendants of an original cycle, every vertex is a representative of exactly one cycle.

We use the representatives to decide where to attach the smaller block's cycle edges to each larger-block cycle. This partitions the cube's edges into anchored products. Each anchored product is a grid with wraparound in both directions, or torus, with one direction subdivided into P-edge segments. The two-wiggle construction splits each product into two simple cycles of the required length M_d. At a shared vertex, each cycle has a neighbor that is absent from the other cycle. Exchanging the two edges to these neighbors opens both cycles into simple paths of the same length, giving the required decomposition.

Higher binary weight, non-Mersenne. Both higher-weight families use a common construction. Write $d = P + L + b$, where $P = 2^r$ and $L = 2^s$ are the largest two powers of two in the binary representation of d, and b is the remaining contribution, a positive odd integer. Split the cube into blocks $G = Q_{2mP}$, $H = Q_{2mL}$, and $J = Q_{2mb}$. Representatives spaced $A = P/L$ edges apart first partition the edges of $G \square H$ into anchored products. On each product, we construct an alternating anchored circuit, using the terminology of the SUAMI work on anchored circuits [6]. This is a specially arranged Euler circuit that uses every edge once, with designated degree-two vertices called marks. Number the intervals between successive marks in Euler order. The two visits to each grid vertex occur in intervals whose indices have opposite parity.

The marks determine where to attach the third block's cycle edges. Within each group descended from an original Hamiltonian cycle, every physical base vertex is marked exactly b times. We pair the marks along each Euler circuit without crossings, then use a perfect-matching decomposition to assign the b corresponding Hamiltonian cycles of J as labels. Every base vertex receives every label exactly once. The noncrossing order lets us arrange the attachments into two connected cycles, and the opposite-parity visits ensure that neither cycle repeats a vertex when the Euler intervals are expanded. Each resulting cycle has length M_d. A fork exchange opens the pair into paths, and subdivision supplies the shorter admissible lengths.

For the non-Mersenne case, we construct the larger block's Hamiltonian cycles in pairs. Each pair has an A-edge path in each cycle joining the same two physical vertices. We reverse one cycle and keep the other forward. Binary addresses let us count the marks even when different pairs have different cyclic position functions. The two orientations give exactly one marked appearance for each of b shifts. We then coordinate the local switches so that the paired cycles exchange marks at the same two physical vertices, preserving this count. The chosen turn pattern also provides the fork needed to open the final cycles.

Higher binary weight, Mersenne. Here $d = 2P - 1$, so $L = P/2$, $A = 2$, and $b = L - 1$.

We use the same assembly, but count the unmarked appearances first. Every base vertex lies inside a subdivided segment in exactly L products. A longer turn pattern and a modified binary address system leave exactly one of those appearances unmarked. The remaining $L-1$ appearances are marked. The address system accounts for a difference between cyclic positions that occurs in the smallest starting cube. As before, reciprocal switches preserve the count, and an isolated turn supplies the fork. We treat $d = 7$ and $d = 15$ explicitly at the boundary of the general construction.

3 Proof

3.1 Reduction to maximal lengths

We first justify the reduction introduced in the overview. Throughout this section, an admissible length for Q_n means a positive integer ℓ satisfying $\ell < 2^n$ and $\ell \mid n2^{n-1}$.

Lemma 1 (Odd-part reduction)**.** *Let $n \geq 2$ be even, and let ℓ be admissible for Q_n. Write $\ell = d\,2^a$, where d is positive and odd, and $a \geq 0$. Let $r \geq 0$ be the unique integer satisfying $2^r \leq d < 2^{r+1}$. Then $d \mid n$ and $n = 2md$ for some positive integer m. Moreover, $r \leq n-1$, and*

$$M_d = d\,2^{n-r-1}$$

is an admissible length divisible by ℓ. It is the largest admissible length with odd part d.

Proof. Since $d \mid \ell$ and $\ell \mid n2^{n-1}$, we have $d \mid n2^{n-1}$. As d is odd, it is coprime to 2^{n-1}, so $d \mid n$. The quotient n/d is positive and even, giving $n = 2md$ for some positive integer m.

From $2^{r+a} \leq d\,2^a = \ell < 2^n$, we obtain $a \leq n-r-1$. In particular, $r \leq n-1$, so M_d has a nonnegative exponent. Since $\ell = d\,2^a$ and $M_d = d\,2^{n-r-1}$, this also proves $\ell \mid M_d$.

The bounds $2^r \leq d < 2^{r+1}$ give

$$M_d < 2^n \leq 2M_d.$$

Moreover,

$$\frac{n2^{n-1}}{M_d} = \frac{n}{d}\,2^r$$

is an integer, so M_d is admissible. Every larger length with odd part d is at least $2M_d$, and therefore fails the strict bound $\ell < 2^n$. Thus M_d is the largest admissible length with odd part d. □

The next lemma explains precisely how a decomposition at a longer length yields one at a shorter length.

Lemma 2 (Subdivision of paths)**.** *Let $M \geq 1$ be an integer. Suppose a finite graph has an edge decomposition into simple paths of length M. If $\ell \geq 1$ and $\ell \mid M$, then it has an edge decomposition into simple paths of length ℓ.*

Proof. Cut each original path into consecutive blocks of ℓ edges. Since $\ell \mid M$, these blocks cover the entire path without any remainder. Each block is a simple path because it is a consecutive portion of an original simple path. Every original edge belongs to exactly one block, and edges from different original paths remain disjoint. The resulting paths therefore partition the graph's edges. They may share vertices at the cuts, as permitted. □

Proposition 3 (Sufficiency of maximal lengths)**.** *Suppose that for every even $n \geq 2$ and every positive odd divisor d of n, the cube Q_n has an edge decomposition into simple paths of length $M_d = d\,2^{n-r-1}$, where $2^r \leq d < 2^{r+1}$. Then every even-dimensional cube has a path decomposition for every admissible length.*

Proof. Fix an even $n \geq 2$ and an admissible ℓ. By Lemma 1, its odd part d divides n, and $\ell \mid M_d$. The assumed maximal-length decomposition for this d, followed by Lemma 2, gives the required decomposition into ℓ-edge simple paths. □

By Proposition 3, it suffices to construct the maximal-length decompositions for each odd divisor d. We organize these constructions using the four families below.

3.2 The four families

Lemma 4 (Four-family classification)**.** *Every positive odd integer d belongs to exactly one of the following families:*

1. $d = 1$.
2. $d = 2^r + 1$ *for some integer* $r \geq 1$.
3. d *has at least three ones in its binary expansion and is not a Mersenne number.*
4. $d = 2^{r+1} - 1$ *for some integer* $r \geq 2$.

Proof. A positive odd integer has a finite binary expansion with a one in its lowest position. If its binary weight is one, that lowest bit is its only one, so $d = 1$. If its binary weight is two, its other one is in some position $r \geq 1$, so $d = 2^r + 1$.

Now suppose the binary weight is at least three, and let r be its highest occupied position. If there is a zero between positions 0 and r, the number is not Mersenne and belongs to the third family. If there is no such zero, every position from 0 through r is one, and

$$d = 1 + 2 + \cdots + 2^r = 2^{r+1} - 1.$$

There are at least three ones, so $r \geq 2$, giving the fourth family.

The first two families have binary weights one and two, while the last two have weight at least three. Within the latter group, a number either has an intervening zero or does not. Thus the families cover all positive odd integers and are pairwise disjoint. □

In particular, this classification applies to every odd divisor d of the dimension. It remains to prove the maximal-length decomposition for each family, uniformly for every positive integer m in $n = 2md$.

3.3 The case $d = 1$

This first family needs only the established Hamiltonian decomposition of even-dimensional cubes.

Proposition 5 (Odd part one)**.** *For every even $n \geq 2$, the edges of Q_n can be partitioned into simple paths of length $M_1 = 2^{n-1}$. Consequently, they can be partitioned into simple paths of length 2^a for every integer a with $0 \leq a \leq n - 1$.*

Proof. For $d = 1$, we have $r = 0$, so $M_1 = 2^{n-1}$. By the Hamiltonian decomposition theorem [2], the edges of Q_n partition into $n/2$ Hamiltonian cycles, each with 2^n edges.

Split each cycle into two consecutive halves of 2^{n-1} edges. Each half is a simple path because a Hamiltonian cycle repeats no vertex before returning to its starting vertex. The halves share their endpoints, have disjoint edge sets, and together cover the cycle. Applying this to every cycle gives the required decomposition.

Finally, every 2^a with $0 \leq a \leq n - 1$ divides 2^{n-1}, so Lemma 2 gives all the shorter lengths claimed. □

3.4 Square factors

We begin with the classical square construction of Ringel, as described by Mollard and Ramras [4, Section 4]. It partitions the edges of the Cartesian product of a Hamiltonian cycle with itself into two Hamiltonian cycles. The product is a grid with wraparound in both coordinates, called a *torus*. In Lemma 6, we restate this construction and record explicit formulas for the cyclic positions and their congruence property.

Lemma 6 (Square factors)**.** *Let C be a cycle of even length $N \geq 4$. Choose a starting vertex and direction, and number its vertices $0, \ldots, N-1$ in cyclic order. A vertex's number in a chosen traversal is its* cyclic position*. The edges of $C \square C$ partition into two Hamiltonian cycles, both starting at $(0,0)$, whose cyclic positions at (i,j) are*

$$I_+(i,j) = Ni + ((i-j) \bmod N), \qquad I_-(i,j) = N((-j) \bmod N) + ((i-j) \bmod N).$$

For every positive divisor D of N, both cyclic positions are congruent to $i-j$ modulo D.

Proof. The first cycle visits columns $i = 0, \ldots, N-1$ in order. In column i, it visits

$$(i,i),\ (i,i-1),\ \ldots,\ (i,i-(N-1)),$$

with coordinates read modulo N, then moves to $(i+1, i+1)$ to start the next column. The second visits rows $j = 0, -1, \ldots, -(N-1)$ in order. In row j, it visits

$$(j,j),\ (j+1,j),\ \ldots,\ (j+(N-1),j),$$

then moves to $(j-1, j-1)$. Both tours visit every vertex exactly once before returning to the start. In each column, the first tour uses all second-coordinate edges except the second tour's row connector. In each row, the second uses all first-coordinate edges except the first tour's column connector. Thus their edges partition the torus.

The displayed cyclic positions count the preceding columns or rows and the position within the current one. Reducing modulo a divisor D of N removes the multiples of N and gives $i-j$ in both cases. □

The agreement modulo divisors of N will let us select equally spaced vertices consistently when we repeat this construction.

We now apply this construction to cubes. The Cartesian product $G \square H$ has vertices (g,h), and an edge moves along one factor while keeping the other coordinate fixed. A pair of q-bit strings forms a $2q$-bit string, so $Q_q \square Q_q = Q_{2q}$. Thus the Cartesian square of a cube has twice its dimension.

If a graph has a Hamiltonian decomposition with cycles E_t, then the tori $E_t \square E_t$ partition the edges of its square. Each edge in the moving coordinate belongs to exactly one cycle, and the fixed coordinate lies on that same spanning cycle. Applying Lemma 6 to each torus doubles the number of Hamiltonian factors. When the graph is a cube, taking its square also doubles its dimension. Here a *factor* is a spanning subgraph. Each of our factors is a Hamiltonian cycle.

We now repeat this construction, keeping track of the cycles at each stage. Start with a Hamiltonian decomposition of Q_{2m} into m cycles. Each application of Lemma 6 doubles the dimension and replaces each cycle by two new cycles. After j squarings, we therefore have $m2^j$ Hamiltonian cycles in Q_{2m2^j}. Each cycle visits every vertex of this cube, so its length is

$$N_j = 2^{2m2^j}.$$

After j squarings, each original cycle has 2^j descendants. We call these descendants a *group*.

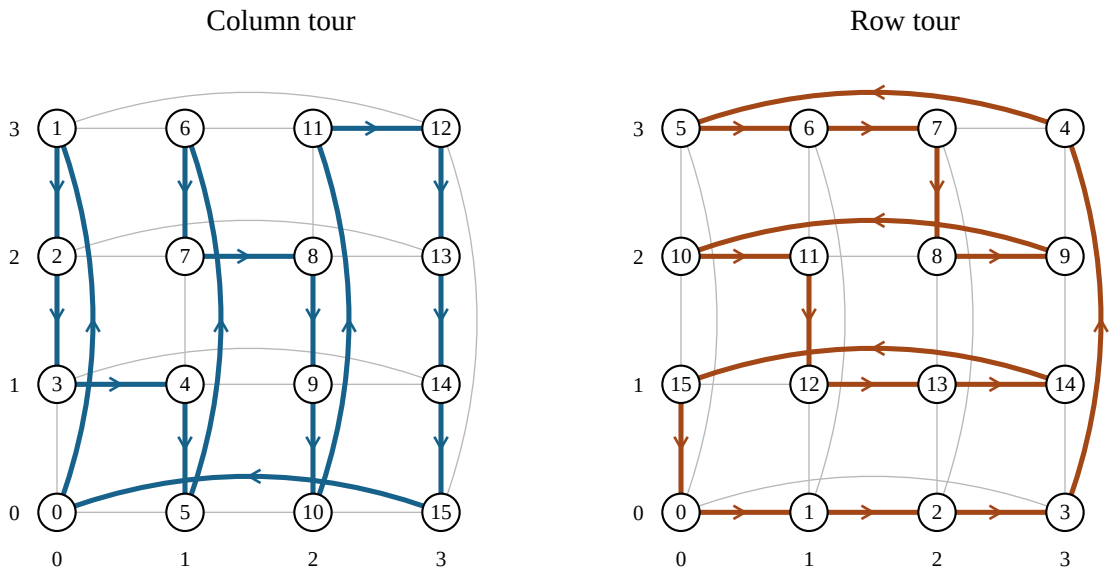


Figure 1: The two tours in Lemma 6 for $C_4 \square C_4$. Both panels use horizontal coordinate i and vertical coordinate j. Numbers inside vertices give cyclic positions, starting at $(0, 0)$, and arrows give the traversal direction. The highlighted edges in the two panels partition the torus edges. Curved edges wrap around the grid. Crossings without a vertex circle do not represent vertices.

Within a group, label a descendant by the choices that produced it. At each squaring, use 0 for the column tour and 1 for the row tour. After j squarings, its choices are $b_0, \ldots, b_{j-1}$, with b_0 recording the first squaring. Place the first choice b_0 in the ones place, the second choice b_1 in the twos place, and so on. Thus the earliest choice occupies the rightmost, lowest bit of the binary address. The resulting number is the descendant's *address*,

$$c = 2^{j-1} b_{j-1} + \cdots + 2b_1 + b_0 = \sum_{u=0}^{j-1} 2^u b_u.$$

Written as a binary number, the address has its bits in the order $b_{j-1} \cdots b_1 b_0$. For example, choosing the row tour first and the column tour second gives $b_0 = 1$ and $b_1 = 0$, so the binary address is 01, which equals 1. Choosing the column tour first and the row tour second gives the binary address 10, which equals 2. Thus the descendants within a group have distinct addresses $0, \ldots, 2^j - 1$.

For each cycle, retain the starting point and direction supplied by the construction. Changing the starting point would change its cyclic positions. Keeping them fixed lets us compare the positions with the addresses in the next lemma.

Every descendant cycle passes through every vertex. We now select equally spaced vertices on each cycle as its *representatives*, assigning every vertex to exactly one descendant within each group. The selection rule compares a vertex's cyclic position with the cycle's address. The next lemma proves that this rule gives every vertex exactly one owner in each group.

Lemma 7 (Representative ownership)**.** *Let $m, r \geq 1$ and $P = 2^r$. After r squarings, the mP Hamiltonian factors of Q_{2mP} form m groups of P descendants, one group for each original cycle. In a group, let $\chi_c(g)$ be the cyclic position of vertex g on the factor with address c. The sets*

$$S_c = \{g : \chi_c(g) \equiv c \pmod{P}\}, \qquad 0 \leq c < P,$$

partition the vertices within that group. Along each factor, consecutive vertices of S_c are separated by P edges.

Proof. The spacing follows from the definition, since P divides the final cycle length. Fix a group and a vertex g. We must show that exactly one address c satisfies $\chi_c(g) \equiv c \pmod{P}$. We will determine its bits successively, beginning with the lowest bit.

Suppose the first j choices $b_0, \ldots, b_{j-1}$ have been fixed, where $0 \le j < r$. They determine a parent cycle of length N_j. This length satisfies

$$2^{j+1} \mid N_j,$$

because $2m2^j \ge j+1$ for $m \ge 1$ and $j \ge 0$. By Lemma 6, choosing either the column tour or the row tour gives the same cyclic position modulo 2^{j+1} at every vertex of the square. Thus the choice of b_j does not affect the lowest $j+1$ position bits at this stage.

This agreement also persists through later squarings. Indeed, Lemma 6 says that, modulo any divisor D of the parent length, a child's position at a vertex (x, y) is the difference of the parent's positions at x and y. If two parent position functions agree modulo D at every vertex, these differences agree too, for either choice of child. All later parent lengths are divisible by $D = 2^{j+1}$, so induction preserves the agreement. Consequently, once the first j choices are fixed, all their final descendants have the same cyclic position at g modulo 2^{j+1}.

In particular, bit j of that final cyclic position is already determined before we choose b_j. Here bit 0 is the lowest binary bit. Set b_j equal to this determined bit. This makes bit j of the address match bit j of the final position, and later choices cannot disturb this match. Starting with $j = 0$ and continuing through $j = r-1$ produces an address whose lowest r bits match those of $\chi_c(g)$. Hence $\chi_c(g) \equiv c \pmod{2^r}$, proving existence.

For uniqueness, any address satisfying this congruence must make the same first choice b_0, since the lowest position bit is independent of all choices. With b_0 fixed, its next choice b_1 is forced by the next position bit. Continuing in this way forces every bit to equal the one we selected above. Thus g belongs to exactly one set S_c. □

3.5 Two binary ones

We now use the representatives from Lemma 7 to decide where to attach the smaller cube's cycle edges to each larger-cube cycle. The unique ownership in each group ensures that each smaller-cube edge is assigned exactly once. Together with the larger-cube cycle edges, these assignments partition the cube's edges into anchored products. We split each anchored product into cycles of the required length and then open them into simple paths. We use the anchored product defined by Axenovich, Offner, and Tompkins [3, Section 3]. The two-wiggle decomposition below is the two-cycle case of their subdivided wiggle construction [3, Section 4.1, Proposition 3]. We give the needed constructions explicitly below.

Lemma 8 (Two wiggles)**.** *Let $K, B \ge 4$ be even, and let $P \ge 2$. Replace every first-coordinate edge of the torus $C_K \square C_B$ by a path of P edges, with distinct internal vertices for distinct edges. The resulting graph has an edge decomposition into two simple cycles, each of length $(P+1)KB/2$. Both cycles contain every original torus vertex, and each internal vertex belongs to exactly one cycle. At any original vertex in row 0, the two neighboring first-coordinate segments belong to opposite cycles.*

Proof. First construct two cycles on the unsubdivided torus. Write $K = 2a$. For each $q = 0, \ldots, a-1$, traverse the columns $2q, 2q+1$ in each row, in that order for even rows and in reverse order for odd rows. In the last row $B-1$, instead visit $(2q+1, B-1)$ followed by $(2q+2, B-1)$, with first

coordinates read modulo K. Between rows move in the second coordinate. The next vertical edge wraps to $(2q+2, 0)$, where the next column pair begins. The last pair returns to the first. Since B is even, all these connecting steps have the required matching first coordinates.

In each ordinary row the column pairs partition the columns. In the last row the shifted pairs do so. Thus this is one Hamiltonian cycle. Translating it one column gives a second Hamiltonian cycle. In ordinary rows, their first-coordinate edges start in complementary column parities. The parities swap in the last row. For second-coordinate edges from (x, j) to $(x, j+1)$, the first cycle uses exactly those with $x + j$ odd, and its translate uses the rest, including the wraparound edges. Hence the two cycles partition the torus edges.

Each cycle takes one first-coordinate edge and one second-coordinate edge per row of each pair, giving $KB/2$ of each kind. Expand its first-coordinate edges to their P-edge paths. Simplicity is preserved because the original vertices were distinct and the inserted internal vertices belong to distinct segments. The length becomes $(P+1)KB/2$. Every internal vertex belongs only to the cycle owning its segment. In row 0, the two segments incident to an original vertex have opposite starting-column parities, so they have opposite cycle owners. □

The edge exchange in Lemma 9 appears in Saranya, Jeevadoss, Nalini, and Chitra [5, Construction 1], who attribute it to Jeevadoss and Muthusamy (2014). We state it separately for use in opening the two cycles.

Lemma 9 (Fork exchange)**.** *Let C, D be edge-disjoint simple cycles of the same length L, sharing a vertex v. Suppose vu is an edge of C, vw is an edge of D, u is absent from D, and w is absent from C. Exchanging these two edges gives two simple L-edge paths that partition $E(C) \cup E(D)$.*

Proof. Removing vu opens C into a simple path from v to u. Adding vw at its endpoint gives a simple path from w to u, since w was absent from C. Removing vw from D and adding vu works in the same way. Each path loses one edge and gains one, and the exchange preserves unique ownership of every edge. □

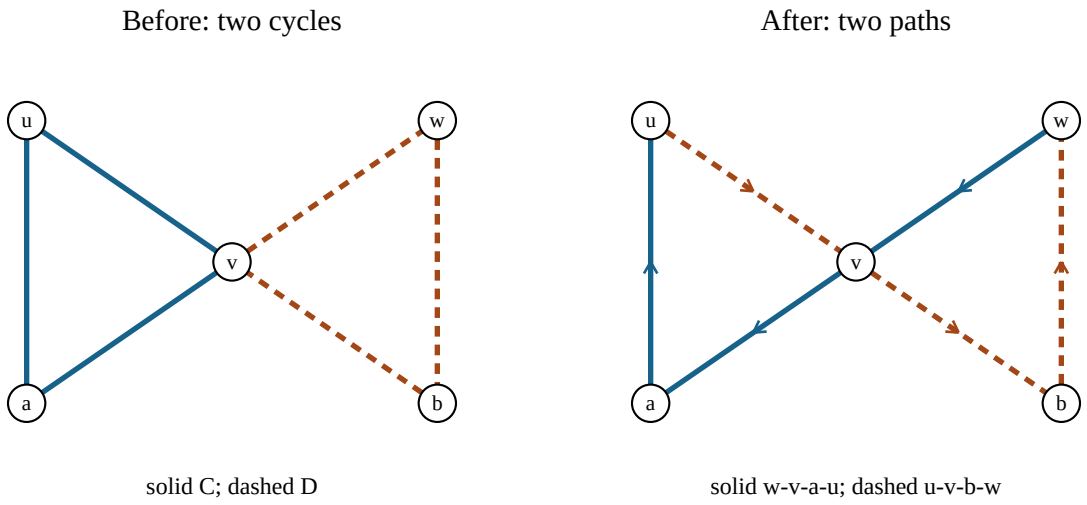


Figure 2: A fork exchange as in Lemma 9, illustrated with two triangles. Exchange ownership of vu and vw. The solid cycle becomes the path w, v, a, u, and the dashed cycle becomes the path u, v, b, w. Each path has three edges, and together they retain every original edge exactly once.

Proposition 10 (Two binary ones)**.** *Let $m, r \geq 1$, $P = 2^r$, $d = P + 1$, and $n = 2md$. The edges of Q_n partition into $2mP$ simple paths, each of length*

$$M_d = (P + 1)2^{n-r-1}.$$

Consequently, every admissible length with odd part d has a path decomposition.

Proof. Split the coordinates into

$$G = Q_{2mP}, \qquad H = Q_{2m}, \qquad Q_n = G \square H.$$

Take a Hamiltonian decomposition $E_1, \ldots, E_m$ of H. Apply r squarings to the same decomposition to obtain the factors of G and their representative sets from Lemma 7.

For each factor C descended from E_t, form its anchored product by including every edge of C at every H vertex, and every edge of E_t only at the representatives of C. For each fixed larger-block vertex g, the vertices (g, h) form a copy of the smaller block. This anchored product includes the cycle E_t in that copy precisely when g is a representative of C.

These products partition the cube's edges. A G-direction edge has a unique factor owner in G. An H-direction edge has a unique owner E_t in H, and its fixed G vertex has a unique representative owner within group t.

Write

$$X = 2^{2mP}, \qquad B = 2^{2m}, \qquad K = X/P = 2^{2mP-r}.$$

In each anchored product, the K representatives along C and the B vertices along E_t form the original vertices of a torus $C_K \square C_B$. Consecutive representatives are joined by P-edge segments, so the product is exactly the subdivided torus in Lemma 8. The required bounds hold. We have $P \geq 2$, $B \geq 4$, and $K \geq 4$, since $2m2^r - r \geq 2$ for $m, r \geq 1$. Both K and B are even.

The two expanded wiggles therefore have common length

$$\frac{(P+1)KB}{2} = \frac{P+1}{2P}\, 2^n = (P+1)2^{n-r-1} = M_d.$$

Choose an original vertex v in row 0. Its nearest neighbors along the two incident first-coordinate segments are distinct internal vertices, because $P \geq 2$. The segments have opposite cycle owners, and each neighbor is absent from the other cycle. Thus Lemma 9 opens the two cycles into simple paths of length M_d while preserving their edge partition.

There are mP anchored products, each giving two paths. Together they partition all edges of Q_n and give the claimed $2mP$ paths. Finally, Lemma 1 and Lemma 2 give every shorter admissible length with odd part d. □

3.6 Construction for higher binary weight

The two remaining families use a common construction to obtain paths of the required length. We develop that construction here and state the inputs it requires. Sections 3.7 and 3.8 supply those inputs for the two families.

We first describe the representative groups needed to form the anchored products. We then construct marked Euler circuits on those products, pair and label their marks, and use the third coordinate block to obtain two simple cycles from each circuit. Finally, we state the conditions under which a fork exchange gives the required path decomposition.

Let the odd integer d have at least three binary ones. Its two largest binary contributions (P and L) and the remaining contribution (b) give

$$d = P + L + b, \qquad P = 2^r > L = 2^s \geq 2, \qquad 1 \leq b < L, \qquad b \text{ odd}.$$

As before, write $n = 2md$, and put

$$G = Q_{2mP}, \qquad H = Q_{2mL}, \qquad J = Q_{2mb}, \qquad A = P/L.$$

Thus $Q_n = (G \square H) \square J$, and A is an even power of two. We call $G \square H$ the base graph. A vertex's *physical parity* is determined by whether the number of ones in its binary label is even or odd. We assign parity 0 for an even number of ones and parity 1 for an odd number of ones. For example, vertex 101 has two ones, so its physical parity is 0. Every cube edge flips exactly one bit, so it changes this parity. A set of marks has balanced physical parity when it contains equally many vertices of each parity.

Write $V(F)$ for the vertex set of a graph F, and $|V(F)|$ for its number of vertices.

We begin with the representative groups needed to partition the base edges. On each Hamiltonian cycle of G, we choose representatives spaced A edges apart. We call these the *old representatives* to distinguish them from the marks used to attach J. Within each original group, we require the P factors to split into L groups of A factors. In each of these smaller groups, the representative sets must partition $V(G)$. The case constructions will supply these groups. Pair these smaller groups with the L corresponding Hamiltonian factors of H. The same ownership argument as in Proposition 10 partitions the base edges into mP anchored products. Each is a torus in which every first-coordinate edge is replaced by a path of A edges.

For the edge count, we use the sizes of the base and third coordinate blocks.

$$C = |V(G \square H)| = 2^{2m(P+L)}, \qquad T = |V(J)| = 2^{2mb}$$

Each anchored product includes one copy of the larger-block cycle at every smaller-block vertex, contributing C edges. Representatives occupy a fraction $1/A$ of the larger-block vertices. Attaching the smaller-block cycle at those representatives contributes another C/A edges. Thus the total number of edges is

$$E = C + C/A = \frac{P + L}{P} C.$$

The last equality uses $A = P/L$.

Next, we prepare each anchored product for attaching edges from J. An Euler circuit is a closed tour using every edge once. It may repeat vertices. We arrange its repeated visits so that they enter different output cycles when the third block is added. The marks specify where to attach those edges. The following definition records the visit conditions we need, using the alternating anchored circuit terminology of the SUAMI construction [6].

Definition 11 (Alternating anchored circuit)**.** An alternating anchored circuit consists of a finite simple graph F, a specified Euler circuit, and $q \geq 4$ distinct marked vertices, where q is even. Every marked vertex has degree two and occurs once in the cyclic Euler order. Every other vertex occurs at most twice. List the marks in Euler order as $z_0, \ldots, z_{q-1}$, and call the route from z_j to z_{j+1} interval j, with indices modulo q. If a vertex occurs twice, its occurrences lie strictly inside intervals of opposite index parity.

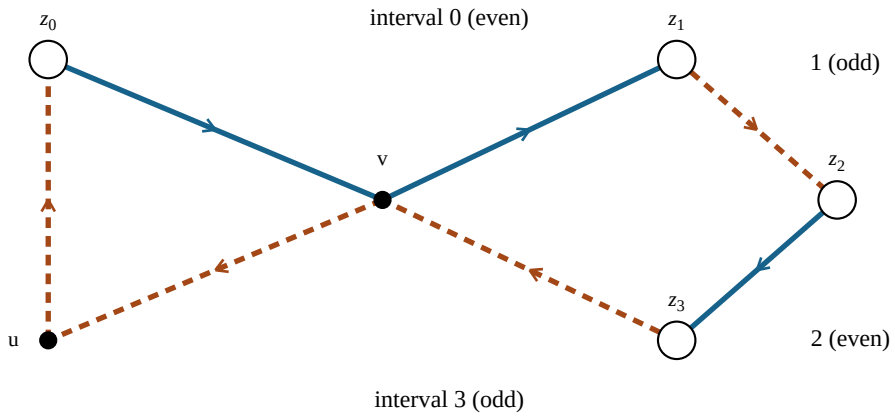


Figure 3: An alternating anchored circuit as in Definition 11. Hollow circles are the four marks. The Euler order is $z_0, v, z_1, z_2, z_3, v, u, z_0$. Every edge is used once, and the two visits to v lie inside interval 0 and interval 3. Solid intervals have even index and dashed intervals have odd index. Interval parity is distinct from physical vertex parity. This example illustrates the definition. The construction below uses subdivided tori.

In our anchored products, the original grid vertices have degree four and occur twice. Inserted vertices have degree two and occur once. The marks will decide where to attach the J-direction edges. The next lemma provides the local Euler construction. Its switches must also be coordinated across products, which the case constructions will arrange.

Lemma 12 (Alternating torus circuits)**.** *Let $K, B \geq 4$ be even, and subdivide every first-coordinate edge of $C_K \square C_B$ into $A \geq 2$ edges. Choose a binary turn pattern t_i, independent of the row, whose period divides K and contains equally many zeros and ones. Suppose*

$$t_{-1} = 1, \qquad t_0 = t_1 = 0.$$

At each grid vertex, pair each first-coordinate edge with a second-coordinate edge according to the turn bit. Changing the turn at $(0, j)$ for $j = 0, \ldots, B-2$ produces one Euler circuit. In every subdivided interval whose endpoint turn bits agree, place a mark at the same offset $z \in \{1, \ldots, A-1\}$, measured in the positive first-coordinate direction. The resulting marked Euler circuit satisfies Definition 11. If h is the number of agreements $t_i = t_{i+1}$ around the original pattern of length K, it has Bh marks. In a cube embedding with even A, its marks have balanced physical parity, both before and after the switches.

Proof. At turn 0, pair the positive first-coordinate edge with the positive second-coordinate edge and pair the two negative edges. At turn 1, use the opposite pairings. Give the second-coordinate edge from (i, j) to $(i, j+1)$ color j mod 2, and give each first-coordinate half-edge the color of its paired edge at the grid endpoint. Within a subdivided interval, the endpoint colors differ exactly when $t_i = t_{i+1}$. In that case, place one color change at the chosen internal mark. Otherwise, keep the color constant throughout. The two transitions at every grid vertex have opposite colors.

To count the baseline tours, put $a_i = (-1)^{t_i}$. Record a state (i, j, σ) after arrival along a second-coordinate edge, where $\sigma \in \{1, -1\}$ is its direction of travel. One first-coordinate traversal followed by one second-coordinate edge gives

$$i' = i - \sigma a_i, \qquad \sigma' = \sigma a_i a_{i'}, \qquad j' = j + \sigma'.$$

The quantity $w = \sigma a_i$ is invariant. Thus the first coordinate advances by the fixed unit $-w$, and after K such steps the second-coordinate displacement is $w \sum_i a_i = 0$. A shorter return is impossible in the first coordinate. There are $2B$ directed tours, paired by reversal into B unoriented tours.

Label a baseline tour by the row where its $w = 1$ orientation arrives in column 0. Since $t_{-1} = 1$ and $t_0 = 0$, the two transitions at $(0, j)$ belong to tours j and $j + 1$. Switching that pairing joins them. The switch at row 0 joins tours 0 and 1. The next joins their union to tour 2, and so on. After the $B - 1$ switches, all tours form one Euler circuit.

Each switched row loses the mark in first-coordinate segment 0 and gains one in segment -1, so its mark count remains h. Hence there are Bh marks in total. The seam gives $h \geq 1$, so $Bh \geq 4$, and Bh is even. A marked degree-two vertex is visited once, and every unmarked inserted vertex is visited once. Each grid vertex is visited twice, once in each local color. Colors change only at marks, so the two visits lie in intervals of opposite parity. This proves the required visit conditions.

For the parity claim, let p be the physical parity of $(0, 0)$. Since every first-coordinate segment has even length A, grid vertex (i, j) has parity $p + j \bmod 2$ in any cube embedding. A mark at offset z therefore has parity $p + j + z \bmod 2$. Every row has h baseline marks, and the even number B of rows gives equal counts of either parity. A switch moves a mark by A first-coordinate edges at the same physical row, preserving its physical parity and hence the balance. □

In a pair of products, the case constructions provide the same physical first-coordinate endpoints, with phases $c, c+A$, and opposite orientations. A switch moves a mark from the phase-c endpoint to the phase-$(c + A)$ endpoint in one product. The corresponding switch moves a mark back between the same vertices in its partner. These transfers preserve the number of marked appearances at each physical vertex, as illustrated in Figure 4. All second-coordinate Hamiltonian cycles are indexed with the same last physical vertex h_*. Switching at every physical H vertex except h_* therefore coordinates the transfers even when the paired products use different Hamiltonian orders.

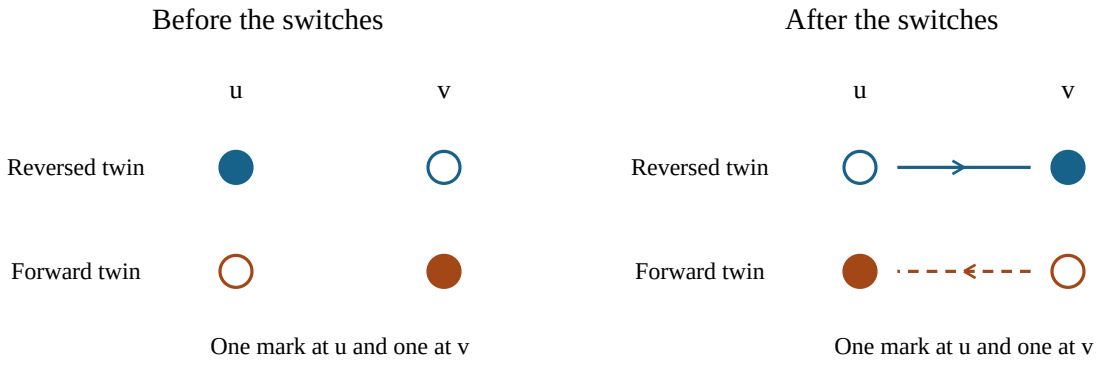


Figure 4: Reciprocal mark transfers in a pair of products. Each column represents the same physical base vertex in both twins, at a fixed H vertex other than h_*. Filled circles are marked appearances. One twin transfers a mark from u to v, while the other transfers a mark from v to u. Each physical vertex retains one mark across the pair. Arrows indicate transfers of marks. They are not graph edges.

We next pair the marks in Euler order, joining opposite physical parities without crossings.

Lemma 13 (Noncrossing pairs)**.** *A cyclic binary word with equally many zeros and ones has a noncrossing perfect matching of its positions, each pair joining opposite symbols.*

Proof. Cut the word at any position. A nonempty balanced linear word has two adjacent opposite symbols, since otherwise it would be constant. Remove such a pair and apply induction to the

remaining balanced word. Restore the removed pair at its gap in the matched word. Since the restored endpoints are adjacent, their pair cannot cross any of the existing pairs. Induction gives a noncrossing matching of the linear order, hence of the cyclic order. □

Lemma 14 (Labels from marked ownership)**.** *Let a family of marked circuits on a common finite bipartite vertex set have balanced physical marked parity in each circuit. Suppose every physical vertex occurs as a mark in exactly $b \geq 1$ circuits, with marks distinct within each circuit. Then the marks admit noncrossing opposite-parity pairings in each circuit and labels $1, \ldots, b$ such that paired marks have the same label and every physical vertex receives every label exactly once.*

Proof. Apply Lemma 13 to the parity word of each circuit's marks. Regard every pair as a separate edge occurrence joining its two physical vertices. The resulting multigraph is bipartite and b-regular. Its edges record pairs of marks and need not be edges of the original graph.

A finite b-regular bipartite multigraph has a perfect matching. Indeed, for a set S on one side, its $b|S|$ incident edges all end in its neighbor set $N(S)$, whose vertices each have degree b. Thus $b|S| \leq b|N(S)|$, giving Hall's condition [7, Theorem 1]. The two sides have equal size by counting all edges from either side, so the matching covers both. Remove it and repeat on the $(b-1)$-regular graph. This partitions the edge occurrences into b perfect matchings. Label each pair by its matching. Every physical vertex is incident with one edge of each label, and the pairings remain noncrossing. □

The next two lemmas develop the circuit-to-cycle attachment described in the SUAMI poster [6], allowing different cycle labels at different marks. We prove this formulation below. The labels will select actual Hamiltonian cycles of J. We first work with an abstract layer for each mark, then expand the connections between layers along the Euler circuit.

Lemma 15 (Two cycles on marked layers)**.** *Let X be a finite vertex set of even size $T \geq 4$, with a balanced bipartition $\pi : X \to \{0, 1\}$. For each label a, let D_a be a Hamiltonian cycle on X whose edges cross this bipartition. Let $q \geq 4$ be even. Assign labels to cyclic layers $0, \ldots, q-1$ so that the layers have a noncrossing perfect matching, with equal labels at the two ends of every pair. Put a copy of D_a in each layer labeled a, and include horizontal edges $(j, x)(j+1, x)$. The resulting graph decomposes into two Hamiltonian cycles.*

Proof. Split each D_a into its alternating perfect matchings $\mathcal{A}_a$ and $\mathcal{B}_a$, regarded as involutions on X. Each involution maps a vertex to its partner in the matching, so applying it twice returns to the original vertex. Color the horizontal edge $(j, x)(j+1, x)$ by $\pi(x) + j + 1 \bmod 2$. Each layer vertex has one horizontal edge of each color.

Choose one matched pair as active and call all other pairs passive. Cut the cyclic layer order just before its opening. For color 0, use $\mathcal{A}_a$ at both ends of every passive pair. At the active opening use $\mathcal{A}_a$, and at its closing use $\mathcal{B}_a$. Color 1 uses the complementary matching at every layer. Each vertex now has one horizontal and one vertical edge of each color.

For color 0, forward horizontal arrivals at layer j have parity $j \bmod 2$. The vertical matching changes that parity, enabling the next forward horizontal step. Color 1 has the opposite arrival and departure sides. Thus a traversal's return map after one lap is the ordered composition of its vertical matching involutions.

Every passive chord lies wholly inside or outside the active chord. In either region, remove an innermost passive chord. Once its nested pairs have been removed, its two identical involutions are adjacent and cancel. Induction cancels every passive pair, without commuting any distinct involutions. The return maps are therefore $\mathcal{B}_a\mathcal{A}_a$ and $\mathcal{A}_a\mathcal{B}_a$ for the active label a.

Because $\mathcal{A}_a \cup \mathcal{B}_a = D_a$ is Hamiltonian, either return map is a single cycle on each bipartition side separately. On the relevant arrival side, its orbit has $T/2$ elements. Following those successive laps visits every arrival vertex in every layer. The intervening vertical matching visits every departure vertex. Each output is consequently one cycle through all qT abstract vertices. The two colors partition every edge. Figure 5 illustrates the construction with four layers. □

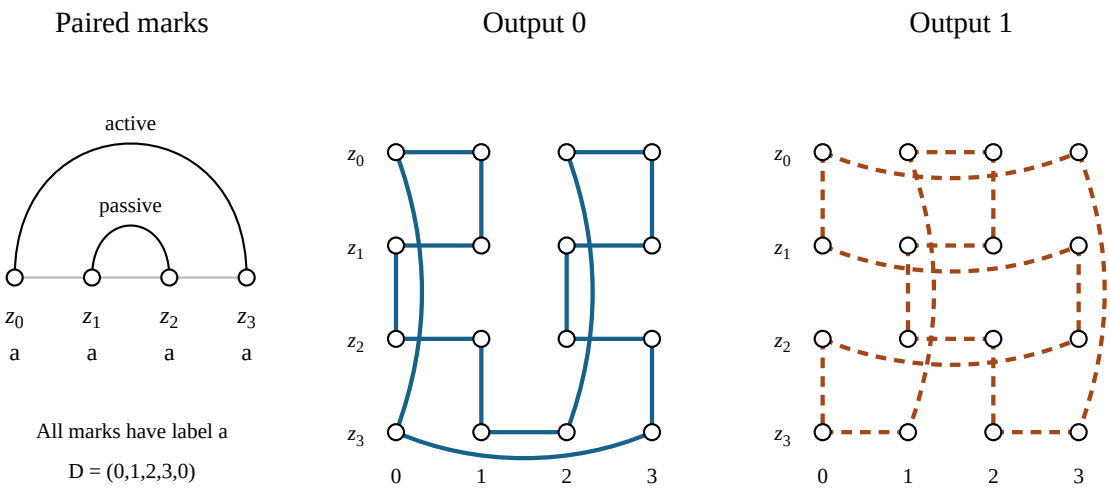


Figure 5: A four-layer example of Lemma 15. Pair (z_0, z_3) is active and pair (z_1, z_2) is passive. All four marks carry the same label a, selecting the cycle $D_a = (0, 1, 2, 3, 0)$. In each output panel, rows are marked layers and columns are vertices of X. Each output is one cycle through all 16 layer vertices, and together their edges partition the layer graph. Curved edges wrap around. Crossings without a vertex circle do not represent vertices. The general lemma permits different labels on different pairs.

Lemma 16 (Expanding the marked layers)**.** *Let F be an alternating anchored circuit with E edges and q marks. Label its marks by Hamiltonian cycles D_a on a common vertex set X of even size $T \geq 4$, all crossing the same balanced bipartition π. Suppose the marks have a noncrossing matching of equally labeled pairs. Form a graph on $V(F) \times X$ by including every F-edge at every $x \in X$ and the edges of D_a at each mark labeled a. This graph decomposes into two simple cycles, each of length $T(E + q)/2$.*

Proof. Apply Lemma 15, with one layer for each mark. Replace each horizontal edge $(j, x)(j+1, x)$ by the actual Euler interval from z_j to z_{j+1} at fixed x.

Marked vertices occur once in the Euler order, and each abstract output visits every marked layer vertex once. An unmarked base vertex occurring once has only one interval occurrence at fixed x. A base vertex occurring twice has its occurrences in opposite-parity intervals. Their colors $\pi(x) + j + 1 \bmod 2$ are opposite, so the two occurrences enter different outputs. These cases show that expansion repeats no physical vertex within an output. Thus the expanded closed tours are simple cycles.

Every base edge occurs in one Euler interval, and each of its copies indexed by X is assigned to exactly one color. At each mark, the two alternating matchings partition its assigned D_a edges. Hence the cycles partition the graph's edges. For each base edge, exactly $T/2$ copies belong to each output, because the bipartition is balanced. Each mark contributes $T/2$ vertical edges to each output. The common length is $TE/2 + Tq/2$. □

The preceding lemma assigns the two visits to a repeated base vertex to different output cycles. It also gives a sufficient condition for the final fork. Suppose a marked vertex, or a base vertex visited twice, has two incident Euler edges leading to distinct unmarked vertices visited once. If those edges lie in intervals of opposite parity, their copies at any fixed $x \in X$ belong to opposite outputs. Their neighboring vertices occur exclusively in those outputs, while the central vertex belongs to both. Thus Lemma 9 opens the two cycles into paths of the same length.

We can now state the common assembly and the three conditions each case must satisfy.

Proposition 17 (Assembly for higher binary weight)**.** *Let $m \geq 1$ and $d = P + L + b$ have the parameters specified above. Suppose the base graph $G \square H$ has an edge partition into alternating anchored circuits $F_{t,c}$, indexed by $1 \leq t \leq m$ and $0 \leq c < P$, with the following properties:*

1. *Each circuit has $E = (P+L)C/P$ edges and $q = bC/P$ marks, with balanced physical marked parity.*
2. *For each fixed t, every physical base vertex is marked in exactly b of the circuits $F_{t,c}$.*
3. *Each circuit has a marked or twice-visited vertex with incident edges in opposite-parity intervals leading to distinct unmarked vertices visited once.*

Then Q_n decomposes into $2mP$ simple paths of length $M_d = d2^{n-r-1}$. Consequently it has a path decomposition for every admissible length with odd part d.

Proof. Take a Hamiltonian decomposition of $J = Q_{2mb}$ and divide its mb cycles into m groups of b, one for each t. Their common vertex set has the balanced physical cube bipartition. Apply Lemma 14 separately to each t-group, using its b Hamiltonian cycles as labels.

For each $F_{t,c}$, form the graph of Lemma 16. These graphs partition $E(Q_n)$. A base-direction edge has its unique circuit owner, independently of the J coordinate. A J-direction edge has a unique Hamiltonian cycle owner in J, and at its fixed base vertex exactly one marked appearance carries that owner's label. Thus every edge is included exactly once.

The fold gives two simple cycles per circuit. The fork condition and Lemma 9 turn them into paths of the same length while preserving their combined edge set. Since $CT = 2^n$ and $P = 2^r$, their common length is

$$\frac{T(E+q)}{2} = \frac{P+L+b}{2P}CT = d2^{n-r-1} = M_d.$$

There are mP circuits and hence $2mP$ paths. Finally, Lemma 1 and Lemma 2 give all shorter admissible lengths with odd part d. □

It remains to construct the circuits and establish these three properties for each of the two higher-weight families.

3.7 The non-Mersenne case

We now construct circuits satisfying the three conditions of Proposition 17. Two Hamiltonian cycles may have different cyclic position functions at the same physical vertex. We therefore keep track of which earlier choices can affect which low binary positions. We also arrange shared endpoints before making the switches that join the Euler circuits. A vertex's phase modulo D means its cyclic position reduced modulo D.

Lemma 18 (Profile cycles and shared paths)**.** *Let R be a power of two, let $D \mid R$, and let A be an even power of two with $2 \le A < D$. Suppose $R \ge 2A$. On the torus $C_R \square C_R$, there are two edge-disjoint Hamiltonian cycles whose cyclic positions, indexed from $(0,0)$, are both congruent to $x+y$ modulo D. Each has a forward A-edge path from $(0,0)$ to $(A/2, A/2)$.*

Moreover, suppose two Hamiltonian cycles on a common vertex set of size M have the same cyclic position function ϕ modulo D, where $D \mid M$ and $M \ge 3D$. If they have forward A-edge paths with the same endpoints and endpoint phases $u, u+A$, then each choice of square tour in Lemma 6 gives a pair of child cycles with shared forward A-edge paths. Their starting phase can be prescribed arbitrarily modulo D.

Proof. Orient both coordinates of the torus positively and put $h = x + y \bmod R$. Choose

$$S = \begin{cases} \{0\}, & A = 2, \\ \{0, \dots, A/2 - 1\} \cup \{A\}, & A \ge 4. \end{cases}$$

The first profile cycle advances x when $h \in S$ and advances y otherwise. The second takes the other positive edge. We call the set of vertices with a fixed value of h a *height fiber*. Both predecessors of a vertex have height $h-1$, so each profile has exactly one incoming and one outgoing edge at every vertex. The profiles partition the torus edges.

In R steps, the first profile translates a height fiber by $(|S|, -|S|)$, and the second translates it by $(-|S|, |S|)$. The number $|S|$ is odd, hence coprime to R. Each translation is therefore one cycle on the R vertices of its height fiber. Each profile is a Hamiltonian cycle of length R^2. Every step increases $x+y$ by one modulo D, proving the cyclic position formula. The first A steps have coordinate words $x^{A/2}y^{A/2}$ and $y^{A/2}x^{A/2}$, so they have the asserted common endpoint and are simple.

For the lifting assertion, the two square-tour position formulas reduce to

$$\psi(X, Y) = \phi(X) - \phi(Y) \pmod{D}.$$

For the column tours, hold X fixed and follow the two parent paths backwards in the second coordinate. To obtain starting phase c, choose $\phi(X) = c + u + A$. A connector interrupts a backwards step only when X is the predecessor of its starting vertex in the relevant parent cycle. For the row tours, hold Y fixed and follow the parent paths forwards in the first coordinate. Choose $\phi(Y) = u - c$. A connector interrupts a step only when Y is the successor of its starting vertex.

In either case, the A starting vertices on each path have distinct phases modulo D. Thus each parent path forbids at most one fixed-coordinate vertex in the required phase class. Three vertices with positions $w, w+D, w+2D$ in either parent cycle are distinct and have that phase, because $M \ge 3D$. At least one avoids both exclusions. With that choice, both child paths use only the prescribed coordinate, are simple, and have the same literal endpoints, at phases $c, c+A$. □

The two paths in the lemma join the same physical vertices. This allows the mark transfers in their descendants to cancel at each endpoint. We call the two profiles *twins*. Subsequent square constructions are called *upper lifts*. We pair their descendants by making the same square-tour choices in both twins. The next lemma selects representatives even when different factors have different cyclic position functions modulo P.

Lemma 19 (Shifted representative ownership)**.** *Let $P = 2^r$, $A = 2^a$ with $1 \le a < r$, and $K = 2^k \le P/4$, where $2mK \ge r$. Starting with one Hamiltonian factor of Q_{2m}, perform k square constructions. Call the resulting K cycles the sources. Apply a profile construction from Lemma 18*

to each source, followed by $h = r-k-1$ further square constructions. There are P final Hamiltonian factors of Q_{2mP}.

Assign the source branch β_j at the jth initial squaring to address position

$$p(j) = \begin{cases} j, & j < a, \\ j+1, & j \geq a. \end{cases}$$

Assign the profile choice ε to position a, and the h upper choices to the remaining positions below r. Write κ for the resulting r-bit address. For every physical vertex g and every $\delta \in \mathbb{Z}/P\mathbb{Z}$, exactly one factor has cyclic position congruent to $\kappa + \delta$ at g. If two shifts agree modulo $2A$, their selected factors have the same profile choice.

Proof. All cyclic positions are indexed from the all-zero cube vertex. At source step j, the parent cycle length is 2^{2m2^j}. We have

$$2m2^j \geq p(j) + 1.$$

For $j = 0$ this is immediate. For $j \geq 1$, use $p(j) + 1 \leq j + 2 \leq 2 \cdot 2^j$. By Lemma 6, both branch choices therefore give the same cyclic position modulo $2^{p(j)+1}$ when the preceding source choices are fixed. Later square constructions preserve this agreement by taking differences. The profile construction preserves it by taking sums.

The source length $R = 2^{2mK}$ is divisible by P. Within a fixed source, both profiles and all their upper descendants consequently have a common cyclic position function χ modulo P, independent of the profile and upper choices. Different sources need not have the same χ.

Decode the address from its lowest position upwards. At a source position $p(j)$, the source prefix already chosen determines $\chi(g) - \delta$ through that position, independently of the remaining choices. Its bit there selects exactly one β_j. At the profile position, the same prefix agreement determines the bit, while the position function is independent of the profile choice. At an upper position, choose the corresponding bit of $\chi(g) - \delta$ directly. This determines one factor, and its completed address satisfies the required congruence. The decoding also proves uniqueness.

If two shifts agree modulo $2A = 2^{a+1}$, the entire decoding through position a is identical. In particular their profile choices agree. □

Proposition 20 (Non-Mersenne odd parts)**.** *Let d be odd, non-Mersenne, and have at least three binary ones. For every $m \geq 1$, with $n = 2md$, the edges of Q_n decompose into $2mP$ simple paths of length M_d, where $P = 2^r$ is the largest binary contribution to d.*

Proof. Write $d = P + L + b$ as in Proposition 17, and put $A = P/L = 2^a$. Either $A \geq 4$, or $A = 2$ and $b \leq L - 3$. Indeed, the excluded combination $A = 2$, $b = L - 1$ is exactly the Mersenne case. Here $r \geq 3$.

Choose $K = 2^k$ to be the least power of two at least $\max(1, r/(2m))$. Since $r \leq 2m2^{r-2}$ for $r \geq 3$, we have $K \leq P/4$. Thus $2mK \geq r$ and $h = r - k - 1 \geq 1$. From each of the m starting Hamiltonian factors of Q_{2m}, construct the P factors and addresses of Lemma 19.

Pair the profile twins with the same source and upper choices. Their addresses are $c, c + A$, where the profile bit of c is zero. The initial profile length is $R^2 \geq P^2 \geq 3P$. Since there is at least one upper lift, Lemma 18 lets us prescribe literal common endpoints of phases $c, c + A$ for every final pair. Reverse the c-factor and retain the $(c + A)$-factor forward.

Choose internal mark offset $z = 2$ when $A \geq 4$, and $z = 1$ when $A = 2$. Index each oriented factor so that its designated endpoint occurs at position z. Its old representatives are the vertices at positions $0 \bmod A$. In terms of its unreversed cyclic position function, their phase classes are

$$\chi(g) \equiv \kappa + z \pmod{A} \quad (\varepsilon = 0), \qquad \chi(g) \equiv \kappa - z \pmod{A} \quad (\varepsilon = 1).$$

Fix all address bits at positions $a, \dots, r-1$ and vary the lower a bits. These A factors form one old representative group. The decoding proof of Lemma 19, restricted to the lower a positions, gives exactly one representative owner at every physical vertex. There are L such groups per starting factor. Assign them to a group of L Hamiltonian factors of H. The resulting anchored products partition the base edges, as in the common assembly.

On each underlying torus, use the period-$2L$ turn pattern

$$t_i = \begin{cases} 0, & 0 \le i \le b, \\ (i-b) \bmod 2, & b < i < L, \end{cases} \qquad t_{i+L} = 1 - t_i.$$

It is balanced, its seam has $t_{-1} = 1$ and $t_0 = t_1 = 0$, and its agreement starts are exactly $0, \dots, b-1 \bmod L$. Its period divides the underlying first-coordinate length $K_0 = |V(G)|/A$. In fact K_0 is divisible by $4L$, since $2mP \ge r + 2$.

Before the switches of Lemma 12, the marks in a reversed factor have phase classes $\kappa - Au$, and those in a forward factor have classes $\kappa + Au$, for $0 \le u < b$. Fix a physical g and one such u. Shifted ownership supplies a unique owner for each of the shifts $-Au$ and $+Au$. These shifts agree modulo $2A$, so their owners have the same profile bit. If that bit is zero, exactly the negative-shift owner contributes a mark. If it is one, exactly the positive-shift owner contributes a mark. Distinct u cannot contribute the same factor twice, because their Au are distinct modulo P. Every physical g therefore has exactly b marked appearances.

Choose a physical vertex h_* of H and put it last in every H-factor's cyclic order. In each product, flip column zero at every row except h_*. Each reversed factor transfers its mark from the phase-c endpoint to the phase-$(c + A)$ endpoint. Its forward twin makes the opposite transfer between the same physical endpoints. Both transfers occur at precisely the physical rows indexed by $V(H) \setminus \{h_*\}$, even if the twins have different H-factors. Thus their combined mark multiplicity is preserved pointwise. Lemma 12 gives one alternating anchored circuit in every product.

There are bK_0/L marks in each row, so each circuit has

$$E = \frac{P + L}{P} C, \qquad q = \frac{bK_0|V(H)|}{L} = \frac{bC}{P}.$$

All its baseline G-marks have the same physical parity. The H-cycle alternates physical parity, with the same number of marks in each row. Each switch preserves that row count and moves a mark by the even distance A. Thus every circuit has balanced physical marked parity. Its mark count is even and at least four.

It remains to check the fork. If $A \ge 4$, a mark at offset 2 has neighbors at offsets 1 and 3. They are distinct unmarked degree-two vertices, visited once, and the mark separates opposite-parity Euler intervals. This remains true after a switch moves the mark to the neighboring segment. If $A = 2$, the bit at column $b + 1$ is isolated between two zero bits in the unflipped row h_*. Both adjacent intervals are unmarked. Their nearest internal vertices are distinct unmarked degree-two vertices. The two incident edges belong to opposite interval parities, since the turn pairs each with an H-edge of opposite row color.

All three inputs to Proposition 17 now hold. It gives the asserted maximal-length paths and, by subdivision, every admissible length with this odd part. □

3.8 The Mersenne case

For $d = 2P - 1$ with $P = 2^r$, we have $L = P/2$, $A = 2$, and $b = L - 1$. Each old representative group consists of two factors with complementary representative sets. Every physical vertex is

therefore an internal subdivision vertex in exactly L products. We will make exactly one of these appearances unmarked. Subtracting it leaves the required $b = L - 1$ marks.

Lemma 21 (Mersenne representative ownership)**.** *Let $r \geq 4$ and $m \geq 1$. Put $P = 2^r$ and $D = 4P$. There is an edge decomposition of $G = Q_{2mP}$ into m groups of P Hamiltonian factors, each arranged into profile-twin pairs with literal shared forward two-edge paths. Orient the first twin backwards and the second forwards. Index the first endpoint at local position 1 in the reversed twin and the second endpoint at local position 1 in the forward twin. For the period-$2P$ turn word $0^{P-1}1^4 01^{P-4}$, the unmarked interior appearances partition $V(G)$ within each group.*

Proof. Choose $K = 2^k$ with

$$4 \leq K \leq P/4, \qquad r + 2 \leq 2mK.$$

Such a choice exists by taking the larger of 4 and the least power of two at least $(r + 2)/(2m)$. For $r \geq 4$, the inequality $r + 2 \leq 2m2^{r-2}$ gives the upper bound. In particular $k \geq 2$ and $h = r - k - 1 \geq 1$.

From each starting Hamiltonian factor, perform k standard squarings. Label the resulting sources by their branch choices $\beta_0, \ldots, \beta_{k-1}$, starting with the earliest choice. Their length $R = 2^{2mK}$ is divisible by D. Apply the $A = 2$ profile of Lemma 18, followed by h upper squarings. This gives P factors per starting factor. Each source has a common final cyclic position function χ_β modulo D, independent of its profile and upper choices.

We assign a numerical phase code e using positions $0, \ldots, r + 1$. Positions 2 and 3 are fixed at zero. The remaining assignments are

$$\begin{aligned}
\text{position } 0 &: \ \beta_0, \\
\text{position } 1 &: \ \varepsilon, \\
\text{position } 4 &: \ \begin{cases} \beta_1, & m \geq 2, \\ \beta_1(1 - \beta_0), & m = 1, \end{cases} \\
\text{position } j + 3 &: \ \beta_j \quad (2 \leq j < k).
\end{aligned}$$

The h upper choices occupy positions $k + 3, \ldots, r + 1$. Here ε selects the profile twin. For a fixed source and upper configuration, write c for the code with $\varepsilon = 0$. The twin codes are $c, c + 2$, with no carry because position 1 of c is zero. The first profile length is $R^2 \geq D^2 \geq 3D$, so the upper lifts let us prescribe shared two-edge endpoints of phases $c, c + 2$ by Lemma 18. Reverse the first twin and keep the second forward. Put their designated endpoints at local position 1. Their old representatives then have phase $e + 1 \bmod 2$.

Use the following balanced turn word, of period $2P$,

$$0^{P-1}\, 1^4\, 0\, 1^{P-4}$$

where an exponent denotes the length of a run. Its disagreement starts are

$$B_0 = \{P - 2, P + 2, P + 3, 2P - 1\}.$$

The unique interior vertex of an agreement segment is marked, and that of a disagreement segment is unmarked. A reversed factor is unmarked at phases $e - 2i$ for $i \in B_0$, and a forward factor at phases $e + 2i$ for $i \in B_0$. Thus, relative to c, one twin pair has the eight unmarked offsets

$$\{0, 2, 2P - 6, 2P - 4, 2P - 2, 2P + 4, 2P + 6, 2P + 8\} \pmod{D}.$$

For $r \geq 4$, reducing these offsets modulo 16 gives each of the eight even residues exactly once.

We now prove that every physical vertex has a unique unmarked owner. Every $\chi_\beta(g)$ has the physical parity of g, since positions start at zero and each cube edge changes parity. This selects β_0 uniquely. Within a fixed β_0, all source position functions agree modulo 16. Indeed, every source step $j \geq 1$ has parent length divisible by 16, and later profile sums and square differences preserve the agreement.

All codes are congruent to $\beta_0 + 2\varepsilon$ mod 16. The eight offsets just listed therefore select exactly one profile choice ε and one disagreement start i. Put $\delta = -2i$ for a reversed factor and $\delta = 2i$ for a forward factor. From here on, subtract the full offset δ modulo D from the cyclic position, including its carries. We must find exactly one factor satisfying $\chi_\beta(g) - \delta = e$.

At source branch 1, the position functions agree modulo 32 when $m \geq 2$. When $m = 1$, the parent length is 16. The square position formulas give instead

$$I_-(x, y) \equiv 17I_+(x, y) \pmod{32}.$$

To see this, their difference is $16(-x - y)$ mod 32, equal to $16I_+(x, y)$ mod 32. Later source steps have parent lengths divisible by 32. Profile sums and upper differences preserve multiplication by 17, so the two sibling position functions satisfy the same relation at the final vertex.

If $m \geq 2$, position 4 of $\chi_\beta(g) - \delta$ selects β_1. The same rule applies when $m = 1$ and $\beta_0 = 0$, because the sibling position functions then agree modulo 32. When $m = 1$ and $\beta_0 = 1$, their values differ by 16 modulo 32, while position 4 of both codes is zero. Exactly one sibling has zero in that position after subtraction, so it is uniquely selected. Numerical codes may coincide across these two sources. Their different physical cyclic positions distinguish them.

For $j \geq 2$, the source step has

$$2m2^j \geq j + 4.$$

With the earlier branches fixed, all remaining position functions agree modulo 2^{j+4}. Position $j + 3$ of the cyclic position after subtraction therefore selects β_j uniquely. Once the source is fixed, the remaining high positions determine all upper choices. The low positions already match by the modulo-16 selection. This proves existence and uniqueness of the full unmarked owner, including its disagreement start, at every physical vertex. □

Proposition 22 (Mersenne odd parts)**.** *Let $d = 2^{r+1} - 1$ with $r \geq 2$, and let $m \geq 1$ and $n = 2md$. Then Q_n decomposes into $2mP$ simple paths of length M_d, where $P = 2^r$. Consequently every admissible length with this odd part has a path decomposition.*

Proof. First suppose $r \geq 4$ and use Lemma 21. Fix every construction choice except β_0 and pair its two values. This is an old representative group, distinct from a profile-twin pair. Its factors have opposite code parity, and both position functions have the physical vertex parity. Their spacing-two representatives therefore partition $V(G)$. There are $L = P/2$ old groups per starting factor. Assign them to the L Hamiltonian factors in the corresponding group of H.

Every physical g has exactly L interior appearances. Exactly one is unmarked by Lemma 21, so precisely $L-1 = b$ are marked. This count holds at every physical H vertex before the switches. The long turn word has the seam required by Lemma 12, and its period $2P = 4L$ divides $K_0 = |V(G)|/2$.

Put the same physical vertex h_* last in every H-factor. Flip column zero in every row except h_*. Each profile-twin pair exchanges marks at its literal phase-c and phase-$(c + 2)$ endpoints, at the same physical rows. The b-fold mark multiplicity is preserved, and each product becomes one alternating anchored circuit.

There are $2P-4$ agreement starts per period $2P$. Each circuit therefore has

$$E=\frac{3C}{2}=\frac{P+L}{P}C, \qquad q=\frac{(L-1)C}{P}.$$

Its marks have balanced physical parity for the same row-count reason as in Proposition 20. At the unflipped row h_*, the zero at column $P+3$ is isolated between ones. Its two neighboring segments are unmarked. Their distinct degree-two internal vertices give the fork required by the assembly. Thus Proposition 17 proves the assertion for $r \geq 4$.

The boundary $r=3$, $d=15$. Here $P=8$, $L=4$, and $D=32$. Perform two standard squarings on each starting Hamiltonian factor. There are four source cycles on Q_{8m}, each of length $R=2^{8m}$, labeled $0,1,2,3$. Sources 0 and 1 share the first branch, as do sources 2 and 3. Their sibling position functions satisfy

$$\chi_1=\lambda\chi_0, \qquad \chi_3=\lambda\chi_2 \pmod{32}, \qquad \lambda=\begin{cases}17, & m=1,\\ 1, & m\geq 2.\end{cases}$$

All four position functions agree modulo 2. The sibling formulas follow from the same length-16 correction just proved, or from parent length divisibility by 32 when $m \geq 2$.

There are no upper lifts in this case. To obtain literal endpoints, use a shifted one-height profile $S=\{c\}$ on each source square, with starts

$$(c_0,c_1,c_2,c_3)=\begin{cases}(0,16,1,1), & m=1,\\ (0,16,1,17), & m\geq 2.\end{cases}$$

From $(0,c)$ the two profile paths are xy and yx, ending at $(1,c+1)$, with phases $c, c+2$. The two profile cycles are Hamiltonian by the single-height translation argument of Lemma 18. Their positions indexed from the all-zero vertex are still congruent to the source-coordinate sum modulo 32, so the sibling relations persist. Reverse the first profile and keep the second forward, with their designated endpoints at local position 1.

Use the long turn word with $P=8$. Relative to c, a profile pair's unmarked classes are

$$U=\{20,12,10,2,14,22,24,0\} \pmod{32}.$$

The sets U and $U+16$ partition the even residues modulo 32. Thus sources 0 and 1 account for every even-parity physical vertex exactly once. When $m=1$, multiplication by 17 fixes even residues, so the same conclusion holds with its sibling correction. For odd residues, sources 2 and 3 give $U+1$ and $U+17$. For $m=1$, the second set is expressed as $U+1$ in source 3's position function, but multiplication by 17 adds 16 to every odd residue and converts it to $U+17$ in source 2's position function. Hence exactly one unmarked appearance exists at every physical vertex.

The two profiles in sources 0 and 1 have odd old representatives. Those in sources 2 and 3 have even old representatives. Pair factors across these two pools to obtain four old groups with complementary representatives. Every physical vertex has four interior appearances and one unmarked appearance, hence three marks. The shared profile endpoints make the switches reciprocal, preserving the mark count at every physical vertex. The common last row supplies the same fork as in the general construction. Together with the balanced marked parity and the circuit sizes, these facts give the three conditions of Proposition 17 for $d=15$.

The boundary $r=2$, $d=7$. Here $P=4$, $L=2$, $b=1$, and $A=2$. Square each starting factor once to obtain two source cycles on Q_{4m}. Their position functions agree modulo 4 by Lemma 6. On source 0's square use the profile $S=\{0\}$, and on source 1's square use $S=\{1\}$. The four

resulting Hamiltonian factors of $G = Q_{8m}$ share a common position function modulo 4. Their literal two-step endpoint pairs have phases $0, 2$ and $1, 3$, respectively. Reverse the first member of each pair and keep its partner forward. Index the first endpoint at local position 1 in the reversed factor and the second endpoint at local position 1 in the forward factor.

The old representatives of both source-0 factors have odd physical parity, and those of both source-1 factors have even physical parity. Cross-source pairing gives two old groups for $H = Q_{4m}$. Use the period-four turn word 0011. Before switching, its four mark classes have phases $0, 2, 1, 3$, so they partition $V(G)$ exactly once. Put the same physical h_* last in both assigned H-cycles and apply the reciprocal switches. The mark partition is preserved, and the products become alternating anchored circuits with $E = 3C/2$ and $q = C/4$.

For the fork, use column 1 in a flipped row. Its turn is zero and its two neighboring turns are one. Both adjacent segments are unmarked, and their distinct internal vertices give the required exclusive neighbors. Physical marked parity is balanced as before. The common assembly gives $8m$ paths of length $7 \cdot 2^{14m-3} = M_7$.

These cases cover every $r \geq 2$. Subdivision supplies the shorter admissible lengths with the same odd part. □

4 Proof of the main theorem

Theorem 23. *For every even integer $n \geq 2$ and every integer $\ell \geq 1$ satisfying $\ell < 2^n$ and $\ell \mid n2^{n-1}$, the edges of Q_n can be partitioned into simple paths, each having exactly ℓ edges. Different paths may share vertices.*

Proof. Fix an even $n \geq 2$ and a positive odd divisor d of n. Since n/d is even, write $n = 2md$ with $m \geq 1$. By Lemma 4, the odd part d belongs to exactly one of the four families. The corresponding maximal-length decomposition is supplied by Proposition 5 when $d = 1$, Proposition 10 when d has two binary ones, Proposition 20 in the higher-weight non-Mersenne case, and Proposition 22 in the higher-weight Mersenne case. Thus, for every positive odd divisor d of n, the cube Q_n has a decomposition into simple paths of length M_d.

Proposition 3 now gives the result. Explicitly, for any admissible ℓ, Lemma 1 gives $\ell \mid M_d$ for its odd part d. Cutting the constructed paths into consecutive ℓ-edge pieces by Lemma 2 preserves simplicity and partitions every edge exactly once. □

5 Consequences and transfers

5.1 Optimal path counts and vertex structure

For even $n \geq 2$, let $p_{\mathrm{eq}}(Q_n)$ denote the smallest number of simple paths in an edge decomposition in which all paths have the same positive length. The edge count $|E(Q_n)| = n2^{n-1}$ and Theorem 23 determine this parameter exactly.

Proposition 24 (Minimum equal-length path count)**.** *For every even $n \geq 2$,*

$$p_{\mathrm{eq}}(Q_n) = \min_{\substack{d \mid n \\ d \text{ positive and odd}}} \frac{n2^{\lfloor \log_2 d \rfloor}}{d}.$$

Proof. A simple path has at most $2^n - 1$ edges, and a decomposition into paths of length ℓ has exactly $n2^{n-1}/\ell$ paths. By Theorem 23, the attainable lengths are precisely the positive divisors

of $n2^{n-1}$ below 2^n. For each positive odd divisor d of n, the largest such length with odd part d is

$$M_d = d2^{n-\lfloor \log_2 d \rfloor - 1}.$$

Every admissible length with odd part d divides M_d, as established in Lemma 1. Thus its path count is at least $n2^{n-1}/M_d = n2^{\lfloor \log_2 d \rfloor}/d$. Each M_d is attained, so taking the minimum gives equality. □

Proposition 25 (Attaining the edge-count bound)**.** *Let $n \geq 2$ be even. Every decomposition of Q_n into simple paths, with lengths allowed to differ, uses at least $n/2 + 1$ paths. An equal-length decomposition attains this bound if and only if*

$$n = 2(2^s - 1) \quad \textit{for some integer } s \geq 1.$$

In these dimensions its 2^s paths have length $(2^s - 1)2^{n-s}$ and are optimal even among decompositions with unequal lengths.

Proof. Each path has at most $2^n - 1$ edges, so the number of paths is at least

$$\left\lceil \frac{n2^{n-1}}{2^n - 1} \right\rceil = \frac{n}{2} + 1.$$

The equality follows from $n/2 < 2^n - 1$ and $n2^{n-1} = (n/2)(2^n - 1) + n/2$. Write $k = n/2 + 1$. If an equal-length decomposition has k paths, then $k \mid (k-1)2^n$. Since k also divides $k2^n$, subtraction gives $k \mid 2^n$, so $k = 2^s$, with $s \geq 1$, and $n = 2(2^s - 1)$.

Conversely, in that dimension $s \leq n$ and $\ell = (2^s - 1)2^{n-s}$ is positive, below 2^n, and divides $n2^{n-1}$. Theorem 23 provides a decomposition with exactly 2^s paths. □

For example, dimensions $2, 6, 14, 30, 62, \ldots$ admit equal-length decompositions attaining the edge-count bound. In other even dimensions, Proposition 24 still gives the exact minimum for equal-length paths. The minimum for paths of unequal lengths remains a separate question.

Proposition 26 (Omitted vertex sets)**.** *Let $n \geq 2$ be even. In any decomposition of Q_n into $k = n/2 + 1$ positive-length simple paths, exactly k vertices lie on every path and are endpoints of two paths. Every other vertex is omitted by exactly one path and is never an endpoint. If the paths have equal length, their omitted sets partition the remaining vertices into k blocks, each of size $2^n/k - 1$.*

Proof. At a vertex v, let i_v, e_v, and a_v count paths on which v is internal, an endpoint, or absent. Then $i_v + e_v + a_v = k$ and $2i_v + e_v = n$, because the paths partition the incident edges. Subtraction yields $e_v + 2a_v = 2$. Thus either $(e_v, a_v) = (2, 0)$ or $(0, 1)$. There are $2k$ endpoint occurrences in total, so the first case occurs at exactly k vertices. Assign every other vertex to the unique path that omits it. This partitions those vertices into omitted sets. With equal length $\ell = (k-1)2^n/k$, each path omits $2^n - (\ell + 1) = 2^n/k - 1$ vertices. □

5.2 Transfers to Cartesian products

A perfect-matching factorization is a partition of the edge set into perfect matchings. The following transfer partitions a larger graph into spanning subgraphs whose components are hypercubes. We then apply the main theorem within each component.

Proposition 27 (Product transfer)**.** *Let $t \geq 2$ be even, and let $G_1, \ldots, G_t$ be finite simple graphs, each admitting a perfect-matching factorization with the same number a of matchings. For every positive integer ℓ satisfying*

$$\ell < 2^t, \qquad \ell \mid t2^{t-1},$$

the Cartesian product $G_1 \square \cdots \square G_t$ has an edge decomposition into simple paths of length ℓ.

Proof. If $a = 0$, the product has no edges, and the empty collection of paths gives a decomposition. Suppose $a \geq 1$. Number each factor's matchings as $M_{i,1}, \ldots, M_{i,a}$. For a fixed c, retain the product edges whose changed coordinate uses an edge of $M_{i,c}$. Each component of this spanning subgraph is a copy of Q_t. Its vertices are obtained by choosing one matching edge in each factor and allowing either endpoint in each coordinate. Every product edge belongs to exactly one color subgraph, because its factor edge belongs to exactly one matching. Apply Theorem 23 to each component. The transferred paths remain simple and partition all product edges. □

Proposition 28 (Graph families)**.** *Let $t \geq 2$ be even. The conclusion of Proposition 27 applies to the following graphs.*

1. *the Hamming graph $H(t, q) = K_q \square \cdots \square K_q$ for every even $q \geq 2$.*
2. *every product $C_{2m_1} \square \cdots \square C_{2m_t}$, where each $m_i \geq 2$.*
3. *every product of t finite simple a-regular bipartite graphs, including products of t copies of $K_{a,a}$.*

Proof. For K_q, label one vertex ∞ and the others by $\mathbb{Z}_{q-1}$. For each $c \in \mathbb{Z}_{q-1}$, pair ∞ with c and pair $c + j$ with $c - j$ for $1 \leq j \leq q/2 - 1$. Since $q - 1$ is odd, every edge not incident with ∞ has a unique midpoint c. These are $q - 1$ perfect matchings partitioning the complete graph.

Every even cycle has a factorization into its two alternating matchings.

Finally, Hall's theorem supplies a perfect matching in every finite a-regular bipartite graph with $a \geq 1$ [7, Theorem 1]. Repeated removal gives a factorization into a matchings. When $a = 0$, there are no edges to decompose. □

The transfer gives decompositions for the lengths inherited from Q_t. A complete classification of possible path lengths in the larger graphs would require further arguments. For example, $H(6, 4)$ and every product of six even cycles of length at least four admit 48-edge path decompositions.

For a product with $N = \prod_i |V(G_i)|$ vertices and common factor degree a, the product degree is at. Its edge count is $atN/2$, so the resulting number of paths is $atN/(2\ell)$. Whenever $\ell \geq at$, this is at most $N/2$. Gallai's conjecture asks for a decomposition of every connected N-vertex graph into at most $\lceil N/2 \rceil$ paths [9, Section 1]. Thus, for connected products, these decompositions meet the conjectured bound. In particular, the product of six even cycles has degree 12, and its 48-edge decomposition uses $N/8$ paths.

The Graham–Häggkvist conjecture asks whether every m-regular bipartite graph decomposes into copies of each fixed m-edge tree. Häggkvist's general-graph conjecture asks for the same conclusion in every $2m$-regular graph [8, Section 1]. The transfer gives path instances of these conjectures for certain Cartesian products.

If $a = 2^s$ with $s \geq 0$ and $ta < 2^t$, then $ta \mid t2^{t-1}$. A product of t a-regular bipartite factors therefore decomposes into paths of length ta, equal to its degree. This is an instance of the bipartite conjecture. A product of t $2a$-regular bipartite factors also admits paths of length ta, now equal to half its degree, giving an instance of the general-graph conjecture.

For example, six 8-regular bipartite factors give 48-edge paths in a 48-regular product. Six 16-regular bipartite factors give 48-edge paths in a 96-regular product. These deductions establish the conjectured decompositions when the fixed tree is a path and the graph belongs to these product families. The conjectures for arbitrary regular graphs and arbitrary trees require further results.

Gunasekara and Devillers study transitive path decompositions, in which a group of graph automorphisms preserves the decomposition and acts transitively on its paths [9, Section 1]. Our transfer guarantees an edge partition into simple paths of the prescribed length. It imposes no symmetry requirement or uniform endpoint count. The matching-based argument therefore supplies decompositions of the stated product graphs, independently of whether particular instances admit other constructions.

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